Tidal Stream Energy

A field note on the MeyGen array, Pentland Firth

Tidal stream turbines sit on the seabed and turn with the flow of the tide, much as wind turbines turn in moving air. Because seawater is about eight hundred times denser than air, a rotor a few metres across can deliver as much power as a far larger wind turbine.

The MeyGen array in the Pentland Firth, between the Scottish mainland and Orkney, began exporting electricity in 2016. Its first four turbines each have a rated output of 1.5 megawatts and together had delivered more than 50 gigawatt-hours to the grid by 2023.

Unlike wind and sunlight, the tides are predictable years ahead: the timing and strength of every flood and ebb follow the orbits of the Moon and the Sun. The difficulty is not forecasting but engineering — corrosion, biofouling and the cost of maintenance at sea.

Maintenance log — turbine 3

14 March: divers found barnacles and kelp on the nacelle. Blades cleaned, anti-fouling coat renewed.

2 April: pitch bearing replaced after a vibration alarm during a spring tide.

19 May: subsea cable connector inspected, no water ingress. Output back to 1.4 MW on the flood.